AGENTIC EMS

A trade EMS for on-chain capital, where the agent's reasoning is a database query away.

Constraint-bounded agents reallocate capital against quantile forecasts, and every decision traces to the projection that justified it.

The volume moved to machines first. The tooling did not.

~30%
of TVL in top-tier DeFi pools is managed by autonomous agents
Chainlink × Ark Invest, May 2026

80%
of DeFi transaction volume is non-human
Fabric Ventures, Mar 2026

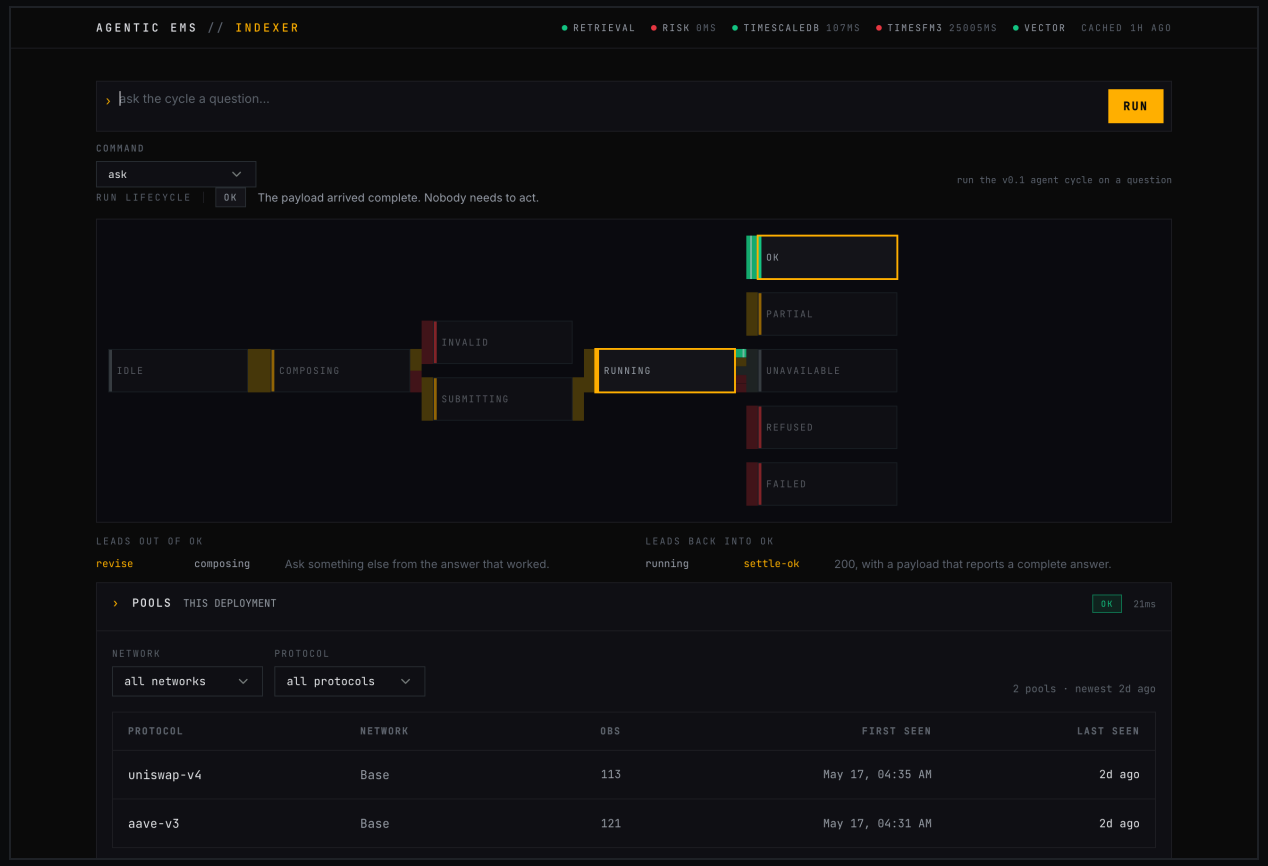
26M
smart wallets, and the standards are only now arriving
Fabric Ventures, Mar 2026

WHAT A DESK HAS IN TRADFI

- Order management, execution management, risk, and a record — four things, one surface.
- Thirty years of hardening against the failure it was bought to prevent.
- Bloomberg is the noun; the desk is the verb.

WHAT A DESK HAS ON-CHAIN

- A block explorer, a multisig, a spreadsheet and a Telegram thread.
- Four windows onto the position. No system holding it.
- **Non-consumption is the incumbent**, and it is free.



The console, live. The pool universe is read from the deployment — an operator never types an address.

- **Ask a question.** An agent runs a five-node cycle: retrieval, yield projection, synthesis, guardrails, readjustment.
- **Forecasts carry a band.** TimesFM-3 issues q10/q50/q90, stored in a forecast ledger — not a single predicted number.
- **Guardrails gate the answer.** A breached constraint forces a re-plan rather than a trade.
- **The record is a row.** Every figure on the surface is a query against what was stored, not a rendering of what the model said.

Yield breaks its band. The position flees. It lands in a vault.

01 · trigger

The band breaks

A strategy's yield falls under its floor, or its volatility breaches its cap. The condition is a forecast with a confidence interval, not a threshold someone typed.

02 · swap

An instruction we wrote

The volatile leg swaps to a stablecoin through a SwapVM program containing `YieldBandFlight` at opcode `0xb3`. The diff against the canonical router is the length of one `if`.

03 · settle

Both legs, one call

`StablecoinRefugeApp` holds no pricing logic. Its whole job is that one call moves both legs back to the maker — removing the window of half-executed risk between the swap and the deposit.

04 · deposit

A real position

The proceeds become shares in a Morpho vault, asserted non-zero in the fork test. The intervention existed for one block and dissolved.

> An intervention, expressed as a hyper-contextualized, intimate experience assembled on the fly, arriving precisely when needed, dissolved once its purpose is served.
>
> FABRIC VENTURES — THE EMERGENCE OF THE MACHINE ECONOMY

TWO AQUA APPS, DELIBERATELY SEPARATE

- **`AgenticEMSSwapVMRouter`** — canonical `SwapVM` + `AquaOpcodes`, plus our instruction. A superset: canonical programs still execute on it at the same opcode, which is why adopting it is not a migration.
- **`StablecoinRefugeApp`** — the atomic exit. Separation is the design, not an accident of packaging.

Every line on this slide is checkable in the repository.

Method and path both named. Reproducible at a pinned fork block, with the evidence committed.

CLAIM	WHERE IT IS CHECKED
29 contract tests, no network	`forge test`
21/21 registry addresses verified against live chain state — on Optimism *and* Polygon	`pnpm run smoke`
`quote(amountOut)` ≡ `swap(amountOut)`	`fork test, re-derived`
The fill landed **exactly on the band** — 3,400 USDC for 1 WETH	`both`
`0xb3` is ours: the same program runs on our router and reverts `UnknownOpcode(0xb3)` on the canonical one	`OpcodeOwnershipTest`
The guard is **inert when unarmed**	`a paired negative test`

Both directions are asserted on the opcode claim. A guard a taker could sidestep by pointing at a different router would be worth nothing; a silent misinterpretation of an upstream opcode would be worse than nothing.

> Will the agents operating in this economy carry verifiable identity, transparent objectives, and genuine accountability? Or will it operate, as Kafka's court operated, from behind a door that recedes the faster you walk toward it?
>
> FABRIC VENTURES — THE EMERGENCE OF THE MACHINE ECONOMY

The difference between a Dickensian and a Kafkaesque outcome is architectural. This is the architecture.

WHAT IS ACTUALLY ENFORCED IN CODE

- **A decision cannot exist without citing its evidence.** `citations: z.array(CitationIdSchema).min(1)` — a reallocation that cannot name the projection or constraint it rests on does not validate.
- **An ungrounded answer degrades rather than passes.** When the model returns a decision that fails validation, the cycle emits an explicit `HOLD` marked `ungrounded` and names the failure in the audit — instead of throwing the synthesis away.
- **Constraints are typed**, and a breach forces a re-plan rather than a trade.
- **The surface states what it cannot verify** — including, today, that the risk store is unreachable and the inference service runs mock.

DATA

Hourly pool metrics across protocols and chains

TimescaleDB · The Graph

FORECAST

TimesFM-3 quantile projections, q10/q50/q90

Forecast ledger, with calibration

DECISION

A five-node agent cycle proposes reallocations

v0.1 LangGraph

RISK

Typed constraints gate the proposal; breaches re-plan

Risk profile · calibration · outcomes

CUSTODY

A Safe owned by an embedded signer

Privy · WalletConnect

SETTLEMENT

Atomic swaps, routed and guarded on-chain

Aqua/SwapVM · Uniswap v4 · 1inch · Morpho

THE TERMINAL

The desk and the console — the surface an operator actually watches. Live today.

THE INFRASTRUCTURE

Forecast, decision, risk, custody and settlement, as one pipeline rather than five scripts.

THE INSTRUMENT

The web3-aligned ETF wrapper. This is the market, and it is the part not yet built.

TOP-DOWN — THE FEE POOL ON MANAGED ON-CHAIN CAPITAL

Crypto hedge fund AUM	$95.0B
Actively managed tokenized / RWA	$26.7B
Addressable capital	**$121.7B**
× 25 bps execution + management	—
TAM	**$304M / yr**

Sources: CoinLaw 2026 · RWA.xyz Jun 2026.

BOTTOM-UP — THE SEGMENT THAT REBALANCES

Crypto hedge funds	400+
Mean AUM	$237M
SAM at 25 bps	**$237M / yr**
SOM at 2% penetration	$4.75M / yr
SOM at 5% penetration	$11.85M / yr

Source: Crypto Fund Research, Mar 2026. The split between funds that trade on-chain and those that hold is `[share of funds that trade on-chain]` — not sourced, so not guessed.

WHAT WE ARE CLAIMING

A **$5–12M ARR** business on the fund segment at 2–5% penetration. Real, and not a decacorn.

WHAT MAKES IT LARGE

The ETF layer, where the capital base is orders of magnitude bigger. `[tokenized fund market, sourced]` — I could not source a credible projection and will not invent one.

THE NUMBER NOT TO QUOTE

$21.5T a year of stablecoin volume — of which **88%** is exchange trading, bots and arbitrage. Flow we do not serve.

Vertical — reported afterwards ↑ structurally required · Horizontal — policy-driven → forecast-driven

	Agentic EMS forecast, and a decision that cannot cite its evidence does not validate
Brava · vaults a codified mandate, reported on afterwards	**Agent platforms** forecast-ish, unconstrained, reasoning unrecorded

THREE KINDS OF ALTERNATIVE

- **Direct — Brava Finance.** The closest competitor on every axis: Safe custody, mandate automation, risk scoring, audit-ready reporting. The difference is **policy versus forecast**, and **reported versus required**.
- **Indirect — vaults, TradFi EMS vendors, institutional DeFi gateways.** Same budget, different instrument. A vault *is* the automation, so an EMS is unnecessary; a TradFi EMS owns the word and not the chain.
- **Non-consumption — a human with a multisig.** Free, understood, never reviewed. **The real incumbent**, and the first slide of this section.

The automated-and-accountable quadrant is **not empty**, and we are not picking different axes to pretend otherwise. What is empty is forecast-driven *and* structurally required — a narrower claim, and a true one.

Any two of these without the third, and you hit a ceiling.

SPECIALISED ACCELERATED COMPUTE

TimesFM-3 issues a quantile band per pool in under a second. Two years ago this was a research paper.

LARGE LANGUAGE MODELS

An agent that can read a constraint set, propose a reallocation, and cite what it read.

PROGRAMMABLE BLOCKCHAINS

Settlement with no intermediary, and a receipt for every action — **the audit trail is native to the substrate.**

ERC-8004

Agent reputation, portable across counterparties. Mainnet, Jan 2026.

EIP-4337

26M smart wallets. An agent with an account rather than a private key in a file.

SETTLEMENT AT SCALE

$27.6T settled on-chain in 2024 — rivalling Visa and Mastercard combined.

Sources: Fabric Ventures, *The Emergence of the Machine Economy* (Mar 2026) · Chainlink × Ark Invest (May 2026).

This list is written by us, about us, and it is already in the repository.

GAP	WHY IT IS THE NEXT THING	KIND
Nothing is committed yet	22 atomic commits planned, each green. Process, not engineering.	`mechanical`
Intent persistence	`exec_intents` carries the `typed_data_hash` binding a signature to the payload an operator reviewed. No repository writes or reads it, so `:id` is a caller-supplied handle.	`next`
Custody caps unenforced	`perTxCapUsdc`, `dailyCapUsdc`, `allowlistOk` exist as schema and nothing consults them. Enforcing needs the store above.	`next`
Position creation unencoded	The entry point is an actions batch whose command bytes could not be retrieved. Left rather than guessed — a wrong byte executes the neighbouring action.	`deferred`
The ETF wrapper	The instrument this is infrastructure for. The market, not yet the product.	`roadmap`

A roadmap that names its own gaps is a stronger signal of relentless execution than a list of features.

DHRUV MALIK — FOUNDER

Six years as a Web3 protocol engineer and smart-contracts auditor, and co-founder of **Extra Labs**, a geospatial 3D reconstruction platform. The execution half of this product is work he has shipped and audited before.

SPECIALISTS

[specialist name] — a former fixed-income trader who went on to run DeFi at a major L2 chain, in discussion as an external specialist. He owns the half a protocol engineer cannot claim alone: how a desk sizes, hedges and defends a position when the band breaks.

[mentor name] — five to six years as a fixed-income trader at Bloomberg, building some of the terminal's most complex components, multi-asset risk synthesis among them. He advises on the risk layer the guardrails are built from.

WHY THIS TEAM

Two halves, two résumés. Protocol engineering — `0xb3` provably ours, Safe-owned custody, an audit trail a schema enforces rather than a report somebody writes — is what the founder has shipped and audited for six years. Fixed-income trading — when to move, what the risk is, how a desk defends a position — is what the specialists bring, one from a desk and one from the machine those desks already trust for risk.

$300K, twelve weeks, and v1 in front of the desks that trade on-chain.

The spend is ordered: build the version that carries the record, then buy the room it is demonstrated in.

1 — BUILD V1

Inference and cloud for the platform's own research, and a minimal salary for a twelve-week base in SF or London.

2 — BUY THE ROOM

The Web3 events worth being at, promoted through Aladdin DAO, with a tutorial series that shows the system working instead of describing it.

3 — OPEN THE MARKET

From what remains: governance proposals to prominent lending protocols and a prediction market. Once v1 is live, sales engineering with the trading firms that run Web3 desks.